

MAJOR CONTRIBUTORS TO THIS ARTICLE



Raghunandan
Vice president-engineering
Kotak Urja



Biju P.C.
Manager-sales and marketing
Emmvee Photovoltaic Power

involved in our manufacturing. But packing is a major concern and we are working on it to reduce the emissions. For this, we have developed recyclable packing designs, which allow all our packing materials to be recycled 100 per cent. Thus, by improving the packing methods and complying to standards, one can drastically reduce the use of wood and paper."

silicon oxide.

"Thus through these processes, other constituent gases also get converted into non-reactive/non-toxic gases with minimal quantity of nitrogen oxides. All these are released into the atmosphere following due industrial and pollution control authority norms. Similarly, during the manufacturing process of solar modules, all fumes emitted are passed through an abatement system and get neutralised in the process."

Using solar panels to produce solar panels. A better strategy would be to use already available solar panels to produce more solar panels. Experts believe that the ecological burden of solar panels can be halved if 100 per cent of the energy required by the factories could be delivered by solar power.

Opting for green manufacturing processes. There are various solar companies that fall under the category of 'zero emission' and 'green' manufacturer, as permitted by the Pollution Board.

Raghunandan informs, "We have inhouse generation of solar power to back up the electricity requirements. We do not have any chemical process

Opting for alternative cell technology. Crystalline cell technology has several alternatives, which are far better than conventional process technologies in terms of greenhouse gas emission. Laser-assisted processes and dry etching are prominent ones. Laser has become an indispensable tool for thin-film PV module production. With most thin-film materials, the layer scribing steps required for monolithic integration cannot be performed with the same flexibility and precision using any other process. This is why laser technology has been a key part of thin-film development for over two decades.

There are several laser-based processes currently used in solar cell manufacturing, or which have a high potential to be implemented in series production in the future. Among these are laser-edge isolation, laser-fired contacts, metal/emitter wrap-through and laser doping. These technologies come at a cost in terms of capital expenditure and also the watts per area produced. Many companies are using these technologies on a commercial scale for volume production.

As more companies begin to develop efficient solar energy collection technologies and higher purity manufacturing systems, the price of solar energy will become more competitive. However, solar energy's economic and environmental benefits could be wiped out if thin-film manufacturers do not design, install and maintain pollution control systems to reduce harmful effluents from the gases released. In such a scenario, we will only be trading one set of problems for another.

Switch to solar energy inevitable

There are always two sides to a coin—while solar energy is considered the safest and most cost-effective renewable energy, its production process is energy-intensive. However, at the cell level the pollution is minimal and at the module level the pollution is negligible or near-zero. According to industry experts, overall, emission per tonne is the lowest among all the energy sources.

Opines Raghunandan, "If civilisation has to move into the twenty-second century, the twenty-first century has to be the century of clean power. Else, we will see devastation in the next 50 years. It is better to invest in solar power before we run out of fossil fuels."

As per recent estimates, coal availability on a global average is expected to last only 60 years. The switch to solar energy is inevitable to save this planet and leave a cleaner world for future generations. ●

The author is an assistant editor at EPY. The article is reprint from January issue of EPY's Electronics Bazaar magazine

KElement
LED TV, LCD TV, 3D TV
pocket Smart TV
Make your Old TV to Internet TV
www.kelement.in
sales@kelement.in

Models
Taupe
Woodbine
Purple orchid
Endive
Bisque
Lilac

Increasing Solutions for 21st Century
(Price Starts from Rs. 4999)

Imagine
innovate
inspire

Trade Enquiry
099943 56789
080560 45640
1800 425 3099